



THE EDGE ON AUTOMATION

Dear Customer:

Congratulations on your selection of Lawrence Equipment machinery. This booklet documents the required information for your Lawrence Equipment **ASPIRATION CONVEY-OR**. It is highly recommended that before attempting to operate the equipment, you read and familiarize yourself with all operating, safety features and maintenance procedures.

Your Lawrence Equipment Service Technician will gladly go over all aspects of your equipment outlining all the required maintenance procedures. Your machine needs these services to maintain the safety, dependability, and performance that has been engineered and built into it. You can always be sure that Lawrence Equipment will be just one phone call away for all your service needs. Calling Lawrence Equipment for all your service needs assures you quality Lawrence Equipment replacement parts.

Keep safety in mind at all times. Before working, cleaning, or maintaining the equipment, it is highly important to lock out all of the energy sources according to OSHA (or local safety) regulations. Only people with the proper qualifications, training, and authority from the management shall work or maintain this equipment.

Please note that due to constant improvement of our equipment, parts of this manual may differ with the actual mechanisms on the machine you have purchased.

Thank you,

LAWRENCE EQUIPMENT



THE EDGE ON AUTOMATION

Technical Manual
01-2024

0AC6010-03 Aspiration Conveyor

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L A W R E N C E E Q U I P M E N T

Product Support Services

AC-601003 **Rev. 1.0-010124**

Subject: **ASPIRATION CONVEYOR**

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This Technical Note applies to all LAWRENCE EQUIPMENT: **ASPIRATION CONVEYOR**

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INTRODUCTION

NOTE: BEFORE WORKING, CLEANING OR MAINTAINING THE EQUIPMENT, IT IS HIGHLY IMPORTANT TO LOCK OUT ALL OF THE ENERGY SOURCES ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

BEFORE ATTEMPTING TO DO ANY WORK, CLEANING OR MAINTAINING THE EQUIPMENT, BE CERTAIN TO SHUT THE POWER OFF BY SWITCHING THE MAIN POWER SUPPLY CIRCUIT BREAKER TO THE OFF POSITION. THE MAIN DISCONNECT SWITCH MUST THEN BE LOCKED OUT AND TAGGED ACCORDING TO OSHA (OR LOCAL SAFETY) REGULATIONS. ONLY PEOPLE WITH THE PROPER QUALIFICATIONS, TRAINING, AND AUTHORITY FROM THE MANAGEMENT SHALL WORK OR MAINTAIN THIS EQUIPMENT.

INTRODUCTION

Congratulations on your selection of Lawrence Equipment **ASPIRATION CONVEYOR** System. This machine is simplicity at its best, incorporating the latest state of the art in its design to offer efficiency in productivity for large and small tortillas.

With the fluctuation of today's market it is difficult to anticipate a day's production. The **ASPIRATION CONVEYOR** System has the versatility that when properly utilized, will allow you production flexibility to meet these varying conditions.

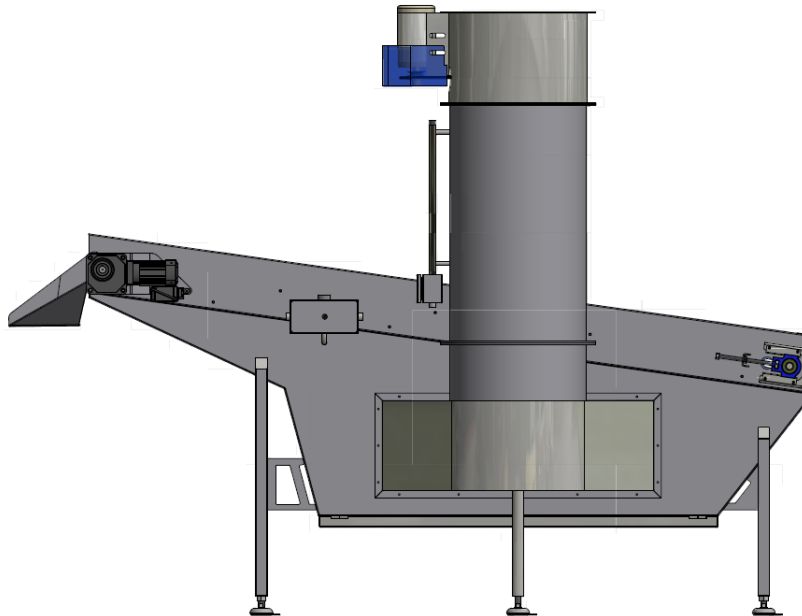
As with all machines dealing with variables, your **ASPIRATION CONVEYOR** System will require some minor adjustments to keep the machine operating to its utmost efficiency.

Simplified maintenance and dependability is designed into the **ASPIRATION CONVEYOR** System, but can only be assured when the machine is operated properly, kept clean, properly adjusted, maintained, and lubricated as outlined in this manual.

The **ASPIRATION CONVEYOR** System has been built with safety in mind. It has safety guards and covers that must remain in place under normal operations and under no circumstances be tampered with or removed. To do so may put your employees' safety in danger. Always disconnect the electrical power source before attempting any cleaning or maintenance on the machine.

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SPECIFICATIONS



- * The Aspiration Conveyor Serial Plate is located on the Fryer Main Control Panel door. The Serial Plate includes the Model, Serial number and Electrical ratings. It is important to know the Model and Serial numbers of the machine when contacting Lawrence Equipment for parts, service, or general ques-

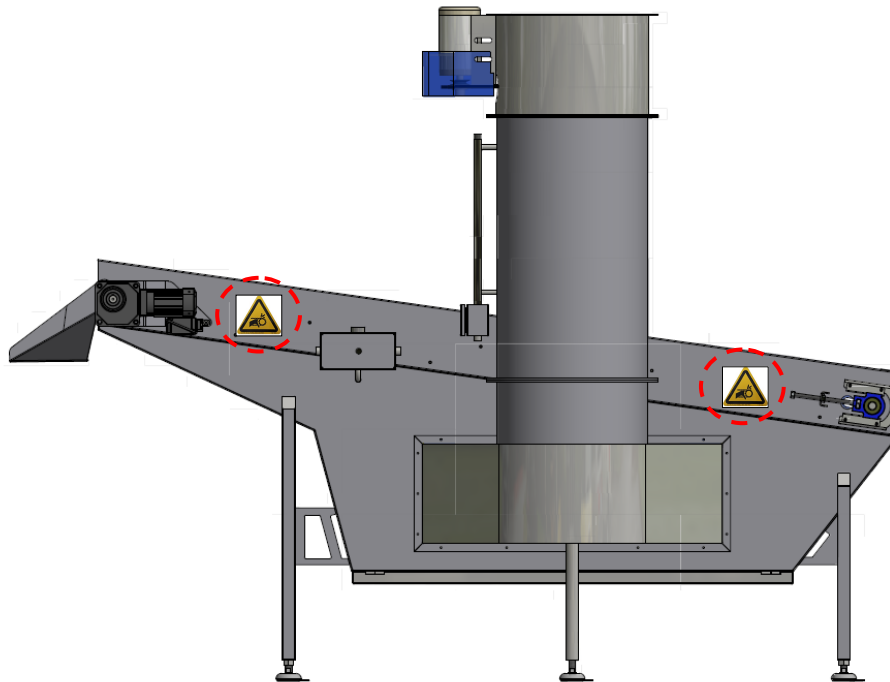
APPLIANCE TYPE		DESIGN ALTD	
MODEL		SERIAL	DATE
VOLTS	AMPS	PH	WIRE HZ H.P.
MAIN GAS PSI		W.C.I.	B.T.U.
MIN & MAX MANIFOLD GAS PRESSURE	/	PURGE TIME	
AIR CFM/P.S.I.		WATER LINE COLD/HOT	
STEAM P.S.I.	LBS/PER HR.	SHORT CIR. CURRENT RATING	

**LAWRENCE
EQUIPMENT**

2034 NORTH PECK ROAD
 SO. EL MONTE, CA 91733
 (626) 442-2894
 (800) 423-4500 Toll Free (except CA)
 FAX (626) 350-5181

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SAFETY LABEL LOCATIONS



Warning Plates/Labels	Description	Serial Plate/Label #
	Warning Hand Entanglement/Chain Drive	NP003-63

Safety Features

- Moving parts, with the exception of all necessary conveying surfaces are covered and guarded.
- DO NOT open doors while running the machine.
- There are emergency stop buttons available. They are usually found in control panels or in remote locations throughout the machines.

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INSTALLATION

Read all instructions before installation. During installation of the machinery, a Lawrence Equipment service technician or engineer must be present to oversee the proper installation and startup of the machine.

Inspection- Check the equipment for damage during shipping. If you see any damage, note it on the Bill of Loading before the driver is allowed to leave, also notify Lawrence Equipment. Check shipping list with number of crates, packages, and parts to assure all items have arrived, including hardware such as nuts bolts, and electrical connections. If any shortage should be found, report them to Lawrence Equipment immediately.

Location and Installation- The machine should be installed on a hard floor with a sufficient load bearing capacity. Install the machine in a place that satisfies the following conditions:

- Smooth and level floor on a solid foundation.
- Adequate ventilation.
- Sufficient space accessible for machine operation and maintenance.
- As per layout (if prepared by Lawrence Equipment).
- If any changes are made during installation consult factory.

Electrical Installation- Connect the electrical supply power using the proper wire size and voltage according to applicable code requirements, and needs of machine. Do Not change unit wiring. Properly trained and licensed personnel should do the wiring.

Motor Oil Level: (Refer to lube chart)- The oil level in motor gearboxes is normally checked and serviced prior to shipping. Always check before startup, as it never hurts to be sure.

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SAFETY NOTICE

- **DO NOT** remove any panels, covers, or door while machine is running.
- **DO NOT** disconnect, remove, or otherwise disable any of the safety interlocks.
- **DO NOT** probe with **anything** into areas where parts or belts are moving (including interiors of panel enclosures, behind chain/sprocket or belt guards).
- **DO NOT** disconnect, jam, or otherwise defeat the operation and purpose of the safety devices such as:
 - Limit switches
 - Door or panel closure switches
 - Safety interlocks
- **DO NOT** place hands or **anything else** into any part of the machine that is in operation or energized and capable of being turned on by accident; or where process material (s) are mixed, fed or other wise worked or conveyed in some manner.
- **DO NOT** allow spilled oil or grease to remain on the floor. Wipe it up before someone slips on it.
- **DO NOT** lift heavy objects alone. Use a forklift or get assistance.
- **DO NOT** push on wrenches. Always try to pull so that if your hand slips you won't injure your knuckles.
- **DO NOT** stand on anything unstable when working above ground level.
- **DO NOT** trust someone else to disconnect the power source. Always check for yourself.
- **DO** wear eye protection when using power tools and when working in the prone position.
- **DO** use ear protection when necessary.
- **DO** plan your job carefully, correctly assemble all components, and double checking all work performed.
- **DO** know where the nearest fire extinguisher is and how to properly use it.
- **DO** learn CPR and general emergency procedures. Encourage your fellow employees to learn them also.
- **DO** carry out work in a logical sequence and make sure that everything is correctly assembled.

OPERATION

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CONTROL PANEL



Panel Descriptions

1. **Main Disconnect Switch**- This switch disconnects the main electrical power to the Chip Fryer, Aspiration Conveyor, Weight Conveyor and Seasoner's Discharge Conveyor when the switch is in the ON position the main control panel will be electrically energized. Always turn the disconnect switch to the off position, and disconnect the main electrical power to the machine before working on and or cleaning the machine.
2. **Safety System Reset**: Pressing this button energizes the safety relay and allows power flow through the control panel. There are three different settings for this Safety Reset Button. A Blue light illuminated means an alarm has faulted and system must be cleared. A blinking light means alarm has been cleared, showing system is ready to be reset. Safety System Reset indicator light off means the control system power is present and system is ready to run.
3. **System Stop**: Activating this button will stop the machine.
4. **Emergency Stops and Remote Location E-Stops**: Pushing these buttons De-Energizes the control system. If the E-Stop button (s) are pushed they will be illuminated red to indicate they are active. If you wish to restart the system locate E-stop active on the control panel by accessing the Alarm Screen then walk to location to pull E-Stop button (s) out Panic Bars should be back into place. Once all E-stops are deactivated the Safety System Energize button should blink blue showing system is ready to be reset.



ALWAYS remember:

1. Prior to running the Aspiration Conveyor, make sure that all safety features are in place and that none have been bypassed. This will not only void any warranties but ensure operator safety.
2. Save all literature and instructions for future use.
3. **DO NOT** remove any placards on this machine.
4. If you are not familiar with the operation of this machine, DO NOT attempt to operate it until you

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CONTROL PANEL (continued)

have received proper training.

5. **DO NOT** spray water on any of the electrical components, as this can cause electrical shorting and injury to the operator.
6. This is a precision machine; follow all instructions on the care and use of this machine. Make sure that you are familiar with the functions of the machine, as this will extend the life of the machine.

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PRE STARTUP CHECKLIST

1. Make sure all access panels, covers and guards are secure and in place.
2. Check all electrical safety interlocks on electrical power panels, doors, guards, and covers are engaged.
3. Make sure all belts, drive chains and sprockets are free of any debris, tools, or any other obstructions that may impair the proper operation of the machine.
4. Check that the main circuit breaker is on.
5. Assure that all sprockets and keys are tight and secure in their respective positions.
6. Assure that all sprockets are aligned properly and are tracking correctly with their respective drive chains.
7. Assure that all Balance Weave wire sprockets are aligned properly with Balance weave wire conveyor.

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ASPIRATION STARTUP

1. Ensure the Safety system on the fryer panel is enabled. Reset the safety if it is not.
2. Ensure the motor disconnect for the blower motor is turned on.
3. To start the conveyor, use the control screen to access the Fryer "Maintenance" screen then select the Aspiration Conveyor "ON".
4. On the same screen, select the Aspiration Blower "ON" to turn on the Blower fan.
4. The Aspiration Conveyor and blower can also be started by selecting the Fryer System Start button, located on Fryer Control Screen. This function will set the complete fryer system into a production ready state.

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ASPIRATION SHUTDOWN

1. To Turn off the Aspiration conveyor, go to the control screen, select the Maintenance screen then select the Aspiration Conveyor "OFF" button.
2. Turn off the Aspiration Conveyor Blower by selecting the blower "OFF" button located on the same screen.
3. The Aspiration Conveyor and Blower can also be shut down by using the Fryer System "OFF" selection when shutting down the entire system. This selection will shut down every function of the fryer system.

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PANEL DESCRIPTIONS



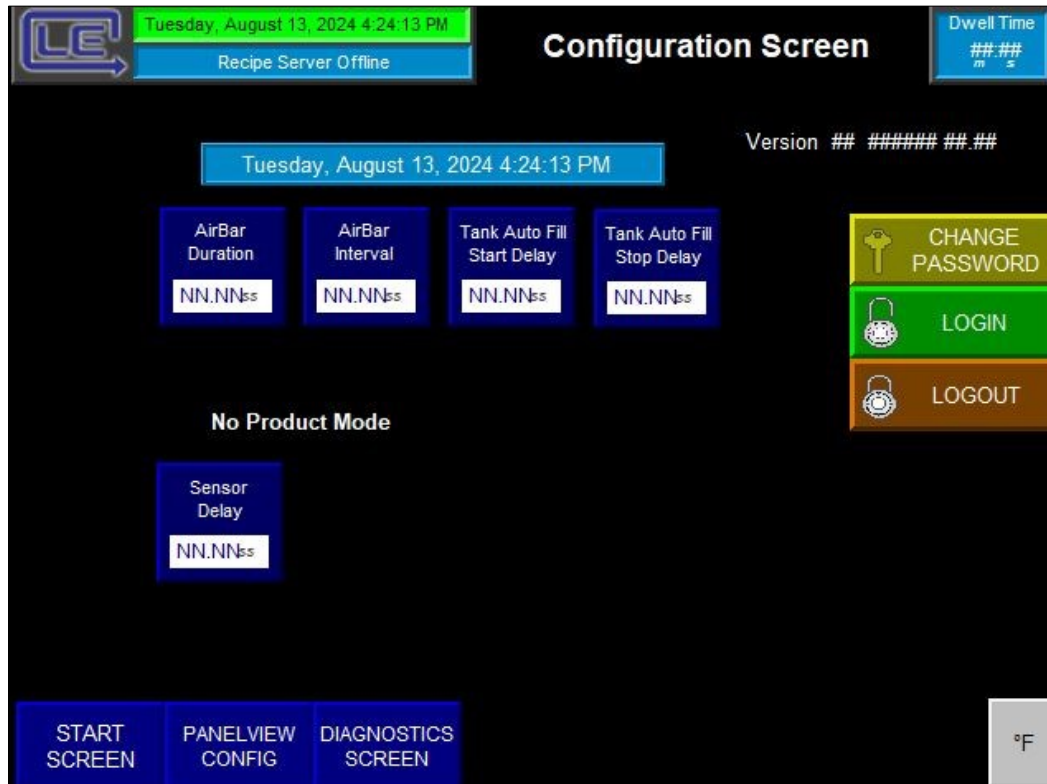
Start Up Screen– When the system is first powered up this is the screen that is displayed.

1. Fryer Screen: This setting allows you to Access the Fryer Screen.
2. Configuration Screen: This Configuration Screen allows you to access setup parameters, however, these screens should only be accessed by Lawrence Equipment Technical personnel.
3. Spanish/English: Allows the user to select English or Spanish.

Note: Some Screens are password protected. Only authorized personnel and Lawrence Equipment should access these screens. Before accessing password protected screens consult Lawrence Equipment, as improper adjustment of any of the values could cause extensive damage to the machine.

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PANEL DESCRIPTIONS (continued)



Configuration Screen- This screen allows you to enter the date (year, month and year), and time (hour, minute and seconds).

1. Change Password: This setting allows the user to change his/her password. The user must first login using their existing username and password.
2. Login: This setting will allow you to login to your account using your user name and password.
3. Logout: This setting allows the user to immediately logout of their account. This is recommended after an adjustment is made so there are no unauthorized adjustments made. **Note:** Session will automatically logout after 5 minutes of inactivity.
4. Air Bar Duration: This numeric entry adjusts how long the Air Bar in the Oil Filter box is activated. A longer value will have the air bar blowing air onto the filter drum longer. A setting of 20 seconds means the air bar will blow air onto the filter drum for 20 seconds. *(Setting 0 ~ 60 sec.)*
5. Air Bar Interval: This numeric entry adjusts the time between Oil Filter box air bar is activations. A longer interval means more time between air activations/Duration. A setting of 20 seconds means a 20 second gap between air bar blows. *(Setting 0 ~ 300 sec.)*
6. Tank Auto Fill Start Delay: This numeric entry adjusts the time that the tank Auto Fill function is activated. This delay starts when the tank low level float is activated by a low fluid level. *(Setting 0 ~ 10min)*

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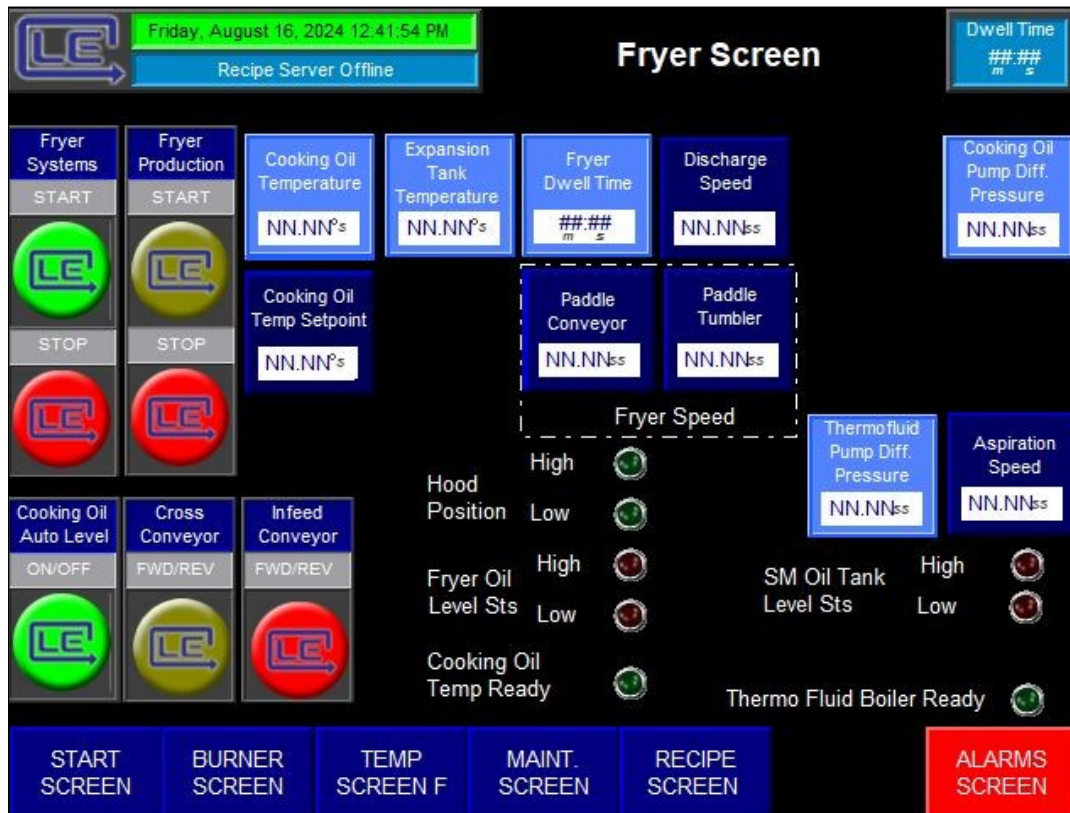
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PANEL DESCRIPTIONS (continued)

7. Tank Auto Fill Stop Delay: This numeric display adjusts the time for the Auto Fill function to stop. Increasing the value allows for more time before the tank fills stops. In example... the Tank auto fill start delays for 5 minutes, when the delay is done the tank fill function starts and the tank auto fill stop delay is activated. With a setting of 5 minutes, the tank will continue to fill for 5 minutes then stop.
8. Product Sensor: This numeric entry adjusts the product detect time. The time delay starts when the product sensor does not detect chips on the fryer infeed conveyor. A setting of 5 seconds means when there are no chips going into fryer for 5 seconds before the system goes into no product mode. *(Setting 0 ~ 10 sec.)*
9. Start Screen: This setting allows the user to access the Start Screen.
10. Panel View Configuration Screen: This setting allows the user to access the Panel View Configuration Screen.
11. Diagnostics Screen: Allows the user to access the Diagnostics Screen.

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PANEL DESCRIPTIONS (continued)



Fryer Screen

1. Fryer System (Start/Stop): This function will turn on all the systems of the fryer, including: Infeed Conveyor, Paddle conveyor, Fryer Discharge conveyor, Cooking Oil Pump, Aspiration conveyor, Aspiration blower, Aspiration blower, Oil Collector Discharge, Filter System and Cooking Oil AutoFill function. This function will not turn on the **Heat Exchanger** systems. (*Please Refer to Heat Exchanger Start-up*).
2. Fryer Production (Start/Stop): This selection will place the system in production mode. This places the cooking oil temperature to the operational run temperature.
3. Cooking Oil Auto Level (On/Off): Select on or off to control the cooking oil level in the kettle automatically. This will allow the system to feed oil very slowly from the storage tank as needed.
4. Cooking Oil Temperature Set point: This is the desired cooking oil temperature. To change it, select the numeric display and input the desired temperature. (*Setting 32°F ~ 450°F*)
5. Expansion Tank Temperature: This indicator displays the current temperature of the Thermo fluid in the Expansion tank. This temperature should not rise above 195 degrees Fahrenheit.
6. Fryer Dwell Time: This indicator displays the current fryer dwell time. This dwell is the actual time the

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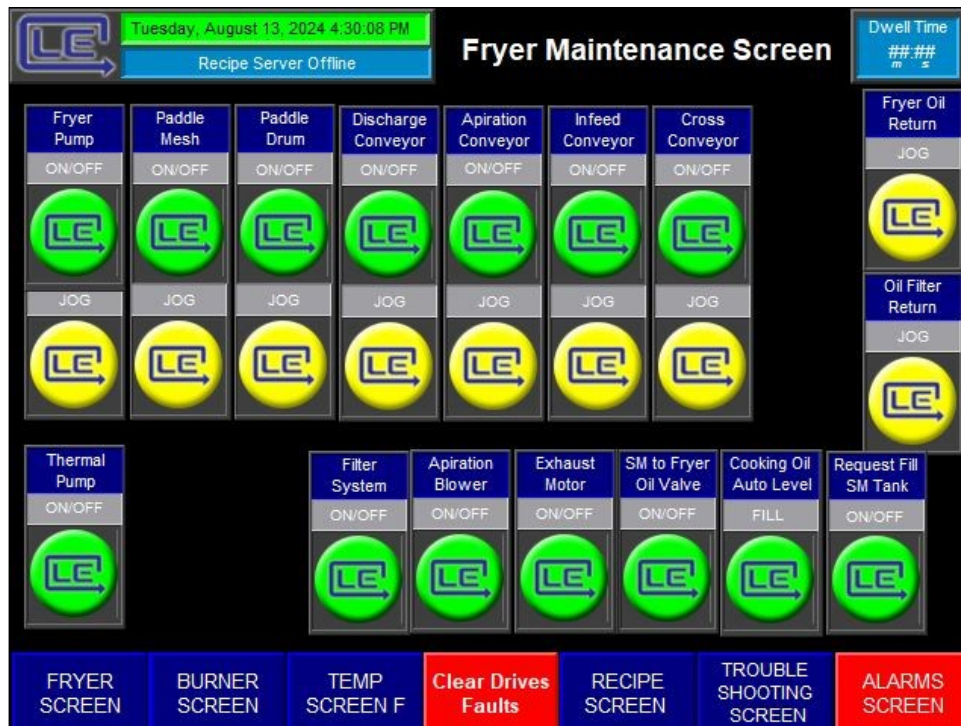
PANEL DESCRIPTIONS (continued)

chips are in the fryer from infeed to discharge.

7. Discharge Speed: This indicator is used to increase or decrease the discharge belt speed. This is indicated in percent of potential run speed. *(Setting 10 ~ 100%)*
8. Paddle Conveyor: This setting adjusts the speed of the Fryer Paddle Conveyor. Increasing this value will increase the speed, decreasing the value lowers the speed. *(Setting 5 ~ 120%)*
9. Paddle Tumbler: This setting adjusts the speed of the fryer paddle conveyor. Increasing this value will increase the speed, decreasing the value lowers the speed. *(Setting 5 ~ 120%)*
10. Cooking Oil Pump Diff. Pressure: Displays pressure differential between the inlet and outlet of the heat exchanger. High input pressure vs. low output pressure can be signs of a obstruction in pipe.
11. Thermo fluid Pump Diff Pressure: This indicator displays the current differential pressure of the Thermo fluid. This value is the pressure difference from the inlet of the thermo fluid heat exchanger to the outlet of the thermo fluid heat exchanger.
12. Aspiration Speed: This setting allows the user to adjust the speed of the Aspiration conveyor. Increasing the speed will lower the total overlap of the product, decreasing the speed will slow the belt down increasing the overlap. *(Setting 10 ~ 100%)*
13. Hood Low Position: Green illumination Indicates the Hood is in the Low Position.
14. Hood High Position: Green illumination indicates the hood is the high position.
15. Fryer Oil Low Level Sts (Status): Illumination indicates that the oil level in the Fryer kettle is low.
16. Fryer Oil High Level Sts Status: Illumination indicates that the oil level in the Fryer kettle is High.
17. Cooking Oil Temperature Ready: Illumination indicates that the Cooking Oil Temperature is ready.
18. Tank Oil Low Level Sts (Status): Indicates that the storage tank oil level is low.
19. Tank Oil High Level Sts (Status): Indicates the storage tank oil level is high.
20. Thermo Fluid Boiler: Green illumination indicates the Thermo Fluid boilers are running.
21. Start Screen: This setting allows access to the Start Screen.
22. Temperature Screen: This setting allows access to the Temperature screen.
23. Maintenance Screen: This setting allows access to the Maintenance Screen.
24. Recipe Screen: This setting allows access to the Recipe Screen.
25. Alarm Screen: This setting allows access to the Alarm Screen.

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PANEL DESCRIPTIONS (continued)



Fryer Maintenance Screen

1. Fryer Pump On/Off: This setting turns the fryer cooking oil pump On and Off.
2. Fryer Pump (Jog): This setting will jog the Fryer oil pump. The pump will run as long as the setting is maintained.
3. Paddle Mesh On/Off: This setting turns On and Off the Fryer Paddle mesh belt conveyor.
4. Paddle Mesh (Jog): This setting will run the mesh paddle conveyor at low speed as long as the setting is maintained.
5. Paddle Drum Off/On: Turns On/Off the Paddle Drum.
6. Paddle Drum (Jog): Momentarily Jogs the Paddle Drum.
7. Discharge Conveyor On/Off: Turns On and Off the Chip Fryer discharge conveyor.
8. Discharge Conveyor (Jog): This setting runs the discharge conveyor as long as the setting is maintained.
9. Aspiration Conveyor On/Off: Turns On/Off the Aspiration Conveyor.
10. Aspiration Conveyor (Jog): Momentarily Jogs the Aspiration Conveyor.
11. Infeed Conveyor On/Off: Turns On/Off the Infeed Conveyor feeding chips into the fryer kettle.

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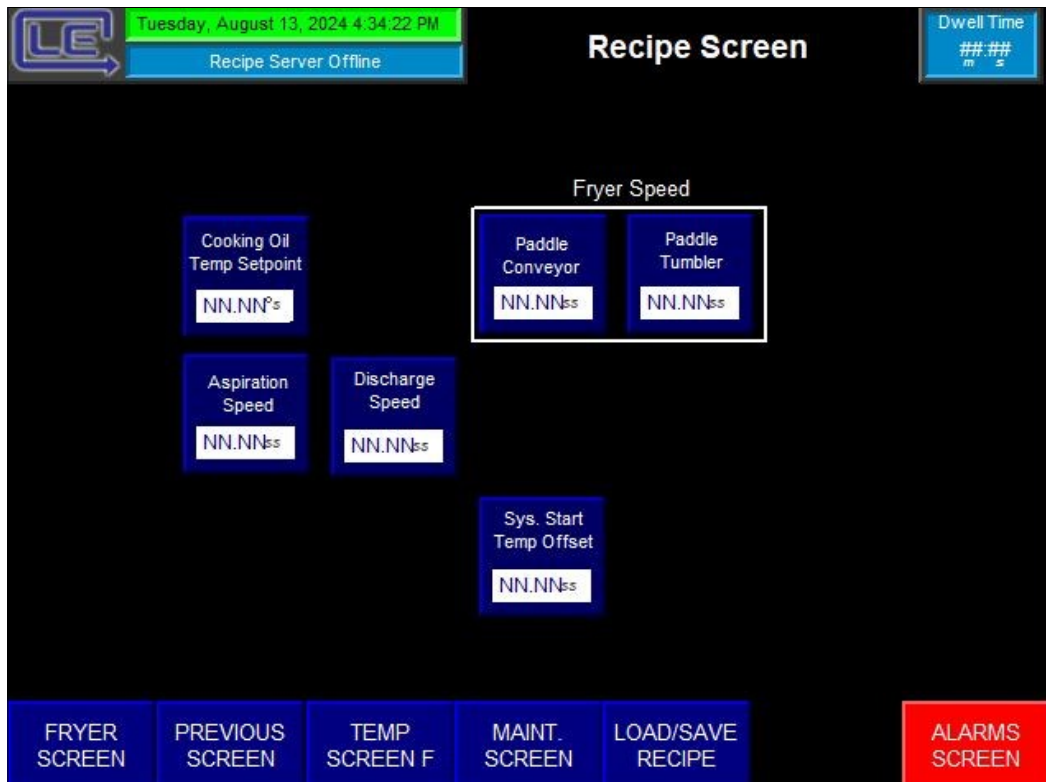
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PANEL DESCRIPTIONS (continued)

12. Infeed Conveyor (Jog): Momentarily Jogs the Infeed Conveyor.
13. Cross Conveyor On/Off: Turns On/Off the Cross Conveyor after the Aspiration Conveyor.
14. Cross Conveyor (Jog): Momentarily Jogs the Cross Conveyor.
15. Fryer Oil Return (Jog): This selection temporarily jogs the fryer oil return. The pump will only stay on as long as the setting is maintained, releasing it stops the pump.
16. Oil Filter Return (Jog): This selection temporarily jogs the Fryer oil filter return pump. The pump will only stay On as long as the setting is maintained, releasing it stops the pump.
17. Thermal Pump On/Off: Turns On/Off the Thermal Fluid Pump on the Boiler module.
18. Filter System On/Off: This setting turns the fryer filter On and Off to remove debris in the Cooking Oil. (This drum filter is located under the heat exchanger).
19. Aspiration Blower On/Off: Turns On/Off the Aspiration Blower.
20. Exhaust Motor On/Off: This setting turns On and Off the fryer exhaust blower. **Note:** This Exhaust motor evacuates the fumes/air from the fryer kettle through the Quick exhaust system.
21. SM to Fryer Oil Valve On/Off: This button will open, if on, the fryer cooking oil valve. Make sure that you use this function for maintenance purposes only, **when this function is active the fryer cooking oil valve will remain open regardless of the oil level in the fryer.**
22. Cooking Oil Auto Level (Fill): This setting will turn on the Auto Fill function, this means that the fryer will control the cooking oil level at it's optimum operational level automatically.
23. Request Fill SM Tank: On/Off: This setting enables the Support Module tank oil request. This will send signal to fresh oil feed pump.
24. Fryer Screen: This setting allows access to the Fryer Screen.
25. Temperature Screen: This setting allows you to access the Temperature Screen.
26. Clear Drive Faults: This setting will clear and reset any drive faults. Any active faults will not clear.
27. Recipe Screen: This setting allows you to access the Recipe Screen.
28. Troubleshooting Screen: This setting allows you to access the Troubleshooting screen.
29. Alarm Screen: This setting allows you to access the Alarms Screen.

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PANEL DESCRIPTIONS (continued)



Recipe Screen

1. Cooking Oil Temperature Set point: This setting allows you to control the temperature of the cooking oil. *(Setting 32 °F ~ 450°F)*
2. Aspiration Speed: This setting adjusts the speed of the Aspiration Conveyor belt. *(Setting 10 ~ 100%)*
3. Discharge Speed: This setting allows the user to adjust the speed of the fryer discharge. Increasing the speed will increase the product flow out of the fryer. *(Setting 10% ~ 100%)*
4. Fryer Speed: This setting allows the user to control the frying time in seconds. *(Setting 5% ~ 120%)*
5. Paddle Conveyor: This setting adjusts the speed of the Paddle Conveyor. *(Setting 5 ~ 120%)*
6. Paddle Tumbler: This setting adjusts the speed of the Paddle Tumbler. *(Setting 5 ~ 120%)*
7. System Start Temperature Offset: This sets the temperature slightly lower than the normal cook temperature for startup. *(Setting 0 ~ 60°F)*
8. Fryer Screen: This setting allows you to access the Fryer Screen.

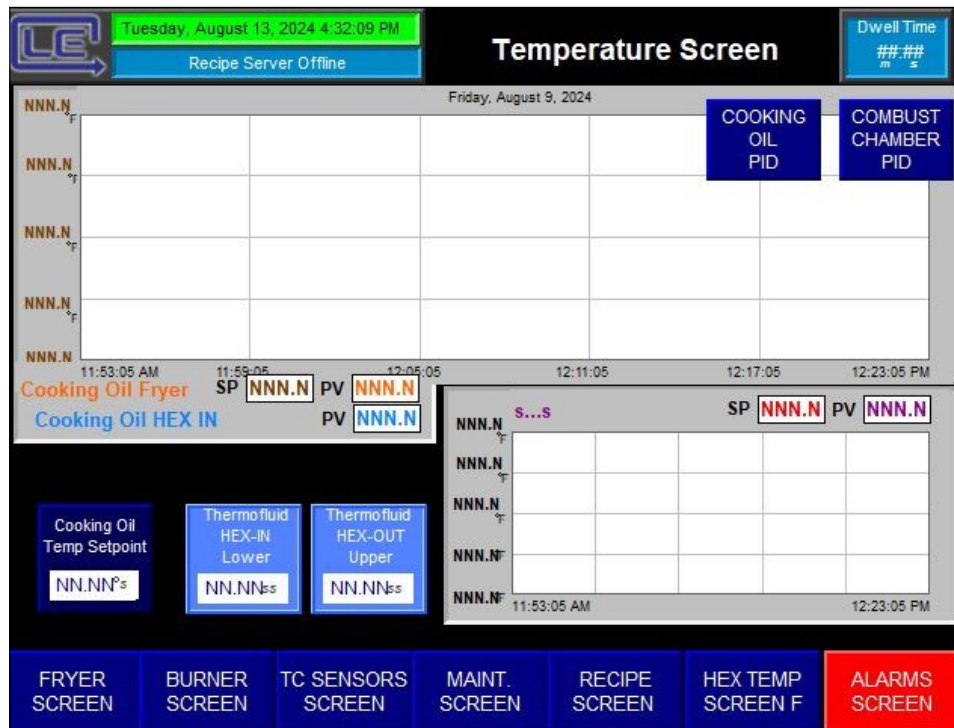
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PANEL DESCRIPTIONS (continued)

- 9. Previous Screen: This setting allows you to access the Previous Screen.
- 10. Temperature Screen: This setting allows you to access the Temperature Screen.
- 11. Maintenance Screen: This setting allows you to access the Maintenance Screen.
- 12. Load/Save Recipe: This setting allows the user to access the Load/Save Recipe Screen.
- 13. Alarm Screen: This setting allows you to access the Alarm Screen.

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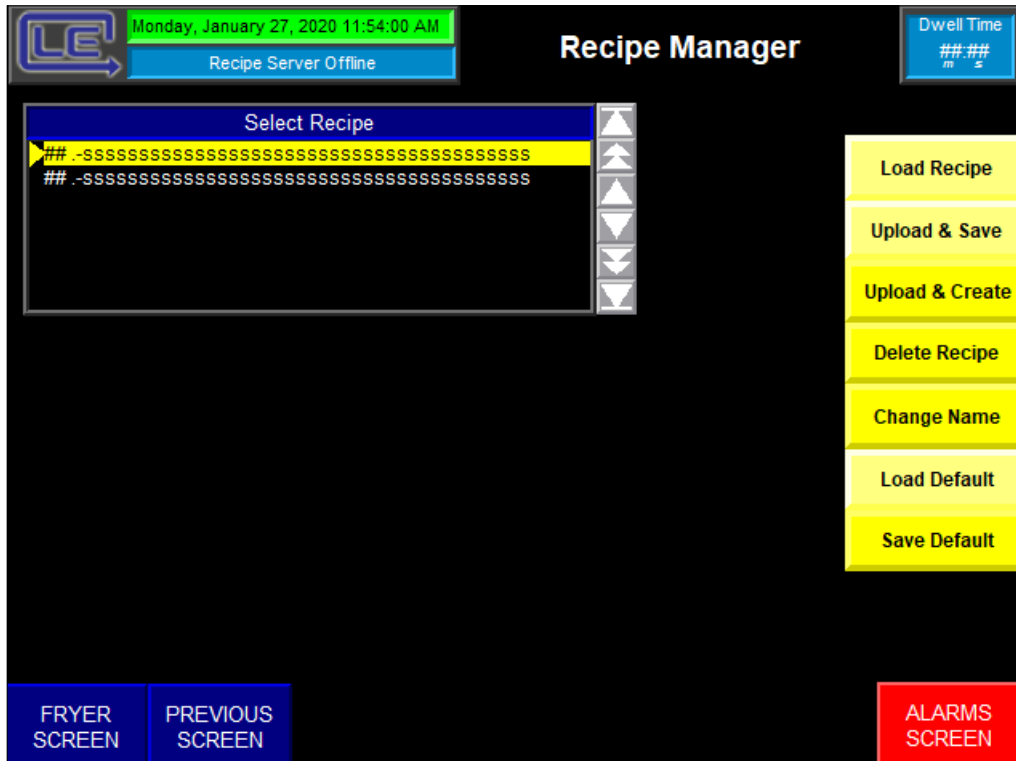


Temperature Screen

1. Cooking Oil PID: Access to the adjustments to the PID program. Technicians only.
2. Combustion Chamber PID: Access to the adjustments to the PID program. Technicians only.
3. Cooking Oil Temperature Set point: This setting allows you to control the temperature set point for the cooking oil. This setting should be set at the optimal temperature for the product being run.
(Setting 32 ~ 450 °F)
4. Thermo fluid Upper Out TC4: This display is the current Thermo fluid temperature measured at the output of the Thermo fluid heat exchanger.
5. Thermo fluid Lower in TC3: This display is the current Thermo fluid temperature measure at the input of the Thermo fluid heat exchanger.
6. Cooking Oil Process Value (PV): This indicator displays the current temperature of the cooking oil
7. Cooking Oil HEX IN (PV): This indicator displays the current temperature of the cooking oil going into the Heat Exchanger.
8. Expansion Tank (PV): This Display is the current temperature of the Thermo Fluid Expansion tank.
9. Fryer Screen: This setting allows the user to access the Fryer Screen.
10. Previous Screen: This setting allows the user to access the Previous Screen.
11. Recipe Screen: This setting allows the user to access the Recipe Screen.
12. Alarm Screen: This setting allows the user to access the Alarm Screen.

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PANEL DESCRIPTIONS (continued)



Recipe Manager

1. Load Recipe: This setting allows the user to load the selected recipe.
2. Upload & Save: This setting allows the user to load the current settings and save the selected recipe.
3. Upload & Create: This setting allows the user to upload the current settings for the new recipe.
4. Delete Recipe: This setting allows the user to delete a recipe.
5. Change Name: This setting allows the user to change recipe name.
6. Load Default: **Note:** This setting should only be accessed by Lawrence Equipment.
7. Save Default: **Note:** This setting should only be accessed by Lawrence Equipment.
8. Fryer Screen: This setting allows the user to access the Fryer Screen.
9. Previous Screen: This setting allows the user to access the Previous Screen.
10. Alarm Screen: This setting allows the user to access the Alarm Screen.

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PANEL DESCRIPTIONS (continued)



Alarms Screen: These screens will appear if any faulted conditions or safety system violations are displayed. These are different levels of alarms. Some stop the production of the line and prevent it from re-starting. Other non-critical alarms do not prevent it from starting but will remain on the screen until the condition is cleared. The alarm setting will flash red when there is an alarm. Alarms should always be addressed as soon as possible to keep the system running to its optimum of safety and performance.

Alarm Status: Displays alarm status setting of all user configurable alarm.

Alarm History: Displays a list of all the alarms that have occurred since last power up.

1. Main Menu Screen: Allows you to access the Main Menu Screen.
2. Previous Screen: Allows you to access the Previous screen.
3. Silence Alarms: Allows you to silence an alarm but will not acknowledge the alarm.
4. Acknowledge Alarms: This setting will allow you to clear any alarms after the condition that created the alarm has been repaired or addressed.
5. Ack All Alarms: Allows you to acknowledge all alarms have been repaired or addressed.
6. Alarm History: Allows you to access the Alarm History Screen.
7. Alarm Status: Allows you to access the Alarm Status Screen.

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ALARM DESCRIPTIONS

Possible Alarm Messages	Description/Action
Aspiration Conveyor Front Right E-Stop Active	This indicates that the emergency stop, located on the front right of the aspiration conveyor is active.
Aspiration Conveyor Front Left E-Stop Active	This indicates that the emergency stop, located on the front left of the aspiration conveyor is active.
Aspiration Conveyor AC Drive Communication Fault	This indicates there was loss of communication between the Aspiration Conveyor AC Drive and the PLC. Make sure that the AC drive is powered ON, check your communication cables and communication module.

ADJUSTMENTS

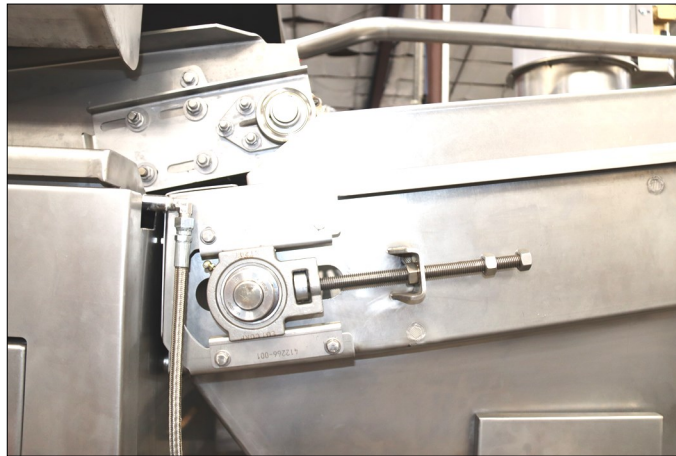
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CONVEYOR TENSIONING

Remove all side access covers. Loosen the idlers to ease chain tension nuts. Tighten each belt conveyor section, or tier, only slightly. Tension-up or tighten belt by loosening tension screw-bar nut, (as mentioned before), and tightening tension screw-bar by turning in a clock-wise direction.

Assure that the belt is tracking straight because of balanced tension, and not being forced by the belt sprockets, which would wear more quickly. After tension adjust, tighten flange bolts, lock nuts, and idlers, do not overly tighten tension.

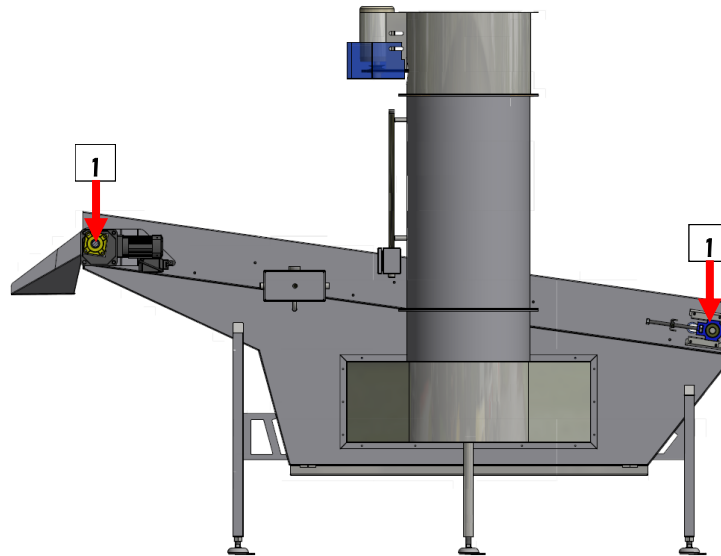


MAINTENANCE SCHEDULE

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LUBRICATION



Item	Part	LE #	Grease/Oil Description	When
1	Bearing	MH027-02	General Purpose grease, food grade	Every 168 hours of operation
2	Bearing	MH027-01	Hi-Temp w/Graphite Grease	NA
3	Chain (oven)	MH027-15	Chain Oil (for Auto Chain Oiler)	Every 72 hours of operation
4	Chain (oven)	MH027-15	Chain Oil (manually applied)	NA
5	Chain (proofer)	MH027-31	Chain Oil (for Auto Chain Oiler)	NA
6	Chain (proofer)	MH027-31	Chain Oil (manually applied)	NA
7	Oil Tank	MH027-06	Hydraulic Oil	A

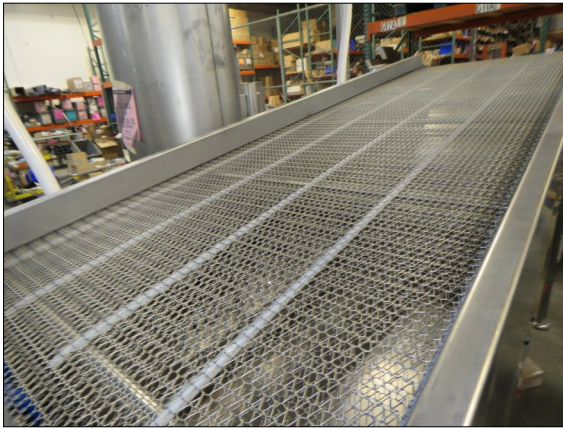
- ★ Grease bearings through zerk fittings provided on bearings. DO NOT force excess amount of grease through the seals. If the bearings become over pressured by too many pumps on a grease gun, the main seals may become damaged or extruded from the bearings. This will cause for the bearing not to hold any grease. Bearing failure will occur. Clean the outsides thoroughly after lubing.
- ★ Motors and speed reducers, etc. are factory set.

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CLEANING

Daily cleaning-

1. Loosen any debris or product deposits from wire belt conveyor.
2. Wash down conveyor with a detergent solution make sure to cover all motors switches before washing conveyor.



Weekly cleaning-

1. Remove panel and guards. With detergent solution cleanse panels and guards. Wipe clean with clean cloth dampened with clean water.
2. Loosen product deposits and debris from wire conveyor and brush away. Cleanse with detergent solution and wipe clean with clean cloth dampened in clean water.
3. Clean the conveyor belt by removing product deposits from the belt surface. Make sure compacted debris is removed from the sprockets, idler wheels and support rails. Pre-rinse the belt and support system with warm water. Apply an appropriate foaming detergent mixture to the belt and support system. Rinse the belt and support system with clean water. After the rinse, inspect the belt and support system components to ensure it is free of any residue. Verify that all cleaning chemical is removed from the conveyor belt, sprockets, idlers, and support rails. Its recommended that pH testing be used as an aid in determining that the belt is free of the detergent. Run the conveyor belt slowly to help dry it and its supports, and remove any pooled water from the floor.
4. Degrease drive chain, motor, etc. With solvent, wipe clean with clean cloth dampened in clean water.
5. Scrub down interior framework, ledges, etc., With detergent solution and rinse clean.
6. Replace cleaned panels, guards, etc.

REPLACING PARTS

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REPLACEMENTS

Another important part to maintaining your machine to its most efficiency is by detecting damaged or worn parts. When following the Maintenance Schedule you have set for your machines, you will be able to inspect specific parts (if not all) of your machine for damage and wear. Do not continue to run machine with damaged parts. Replace faulty parts before a bigger problem occurs.

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BALANCE WEAVE BELT (Assembly Instructions)

Installing Belt

- * The is no top or bottom side to belt either can be up.
- * Direction of travel—insure the spiral leads the crimped connector to which it is welded.
- * If replacing an old belt, splice leading spiral of the new belt to trailing spiral of old belt. *See Splicing belts ends together.*
- * If there is no old belt on conveyor, pull belt through conveyor while maintaining even tension across belt's width.
- * Before applying tension, insure all spirals are seated into the connectors and lie flat.

Splicing Belt ends Together

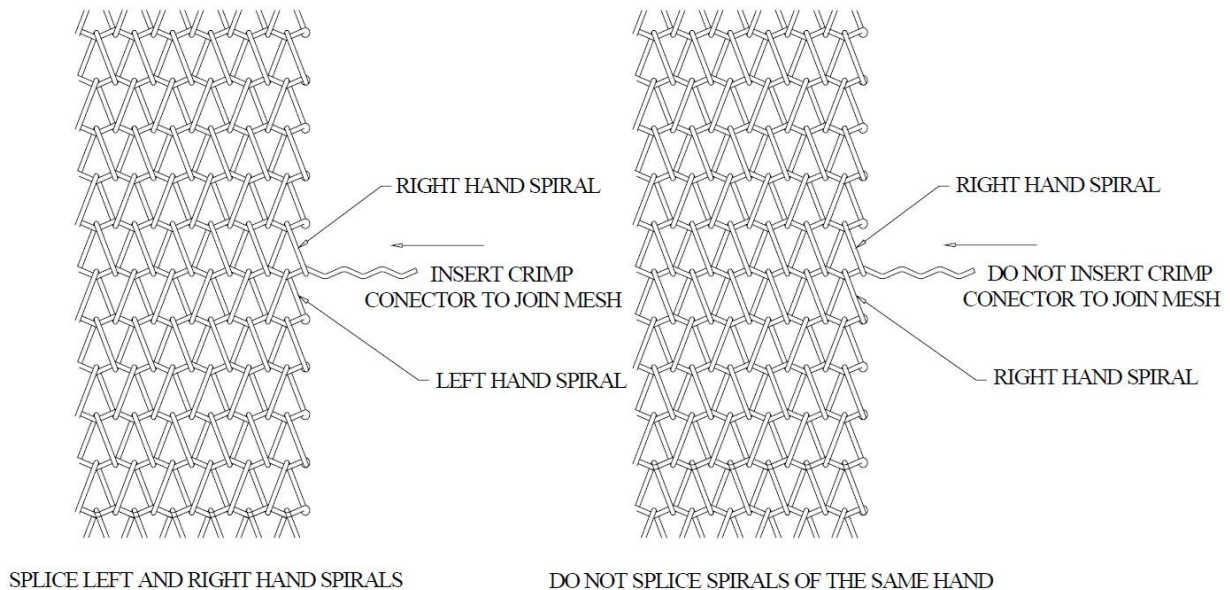
1. Bring the two ends together with the belt edges in line.
2. Mesh the end spirals together permitting insertion of crimped connecting wires. Insure spirals being joined are of opposite weave—one right hand, the other left hand.
3. Insert crimped connector and cut the connector such that approximately 1/16 inch [1.6mm] overhangs the end spirals.
4. Apply a simple wed to fasten connector to end spirals.

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BALANCE WEAVE BELT (Assembly Instructions)

Removing Stretch from Belts

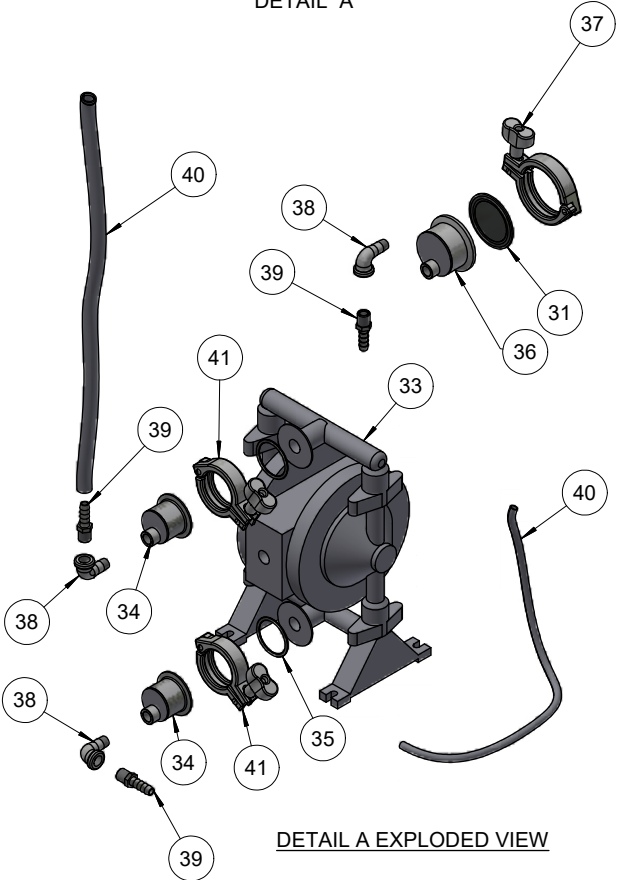
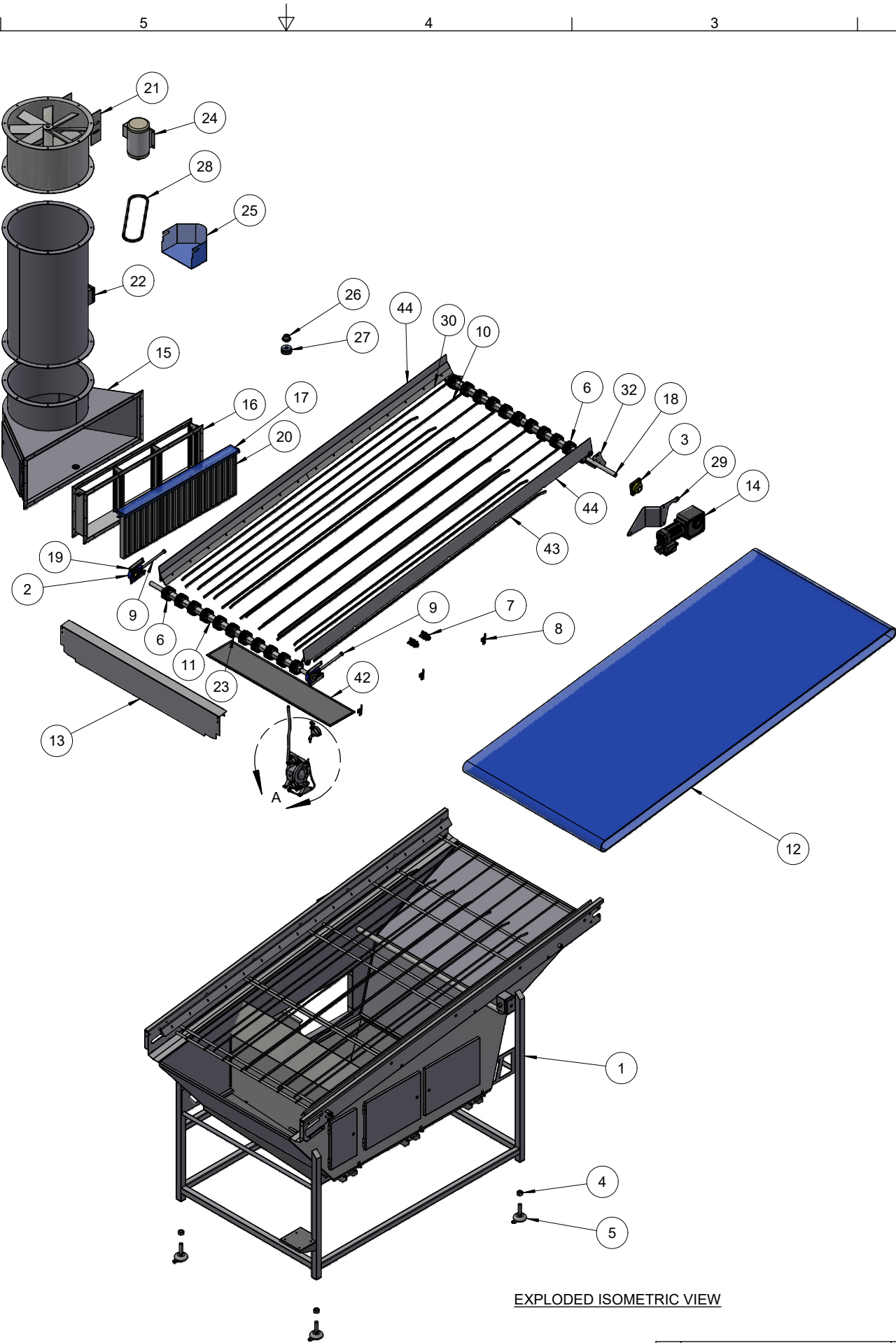
1. Belt length will need to be removed when the take-up mechanism is near its maximum stroke.
2. Identify a section of the belt to be removed. Insure end spirals of section to be removed are of opposite weave.
3. Remove as much belt as possible, however insure enough belt remains to easily splice ends together.
4. On one belt edge, cut at the weld junction leaving the weld on the crimped connector. On other belt edge, cut at the weld junction leaving the weld on the end spiral. Remove the connector, pulling on the connector's end containing the weld.
5. See "Splicing Belt Ends Together" to reattach belt.



REFERENCE DRAWINGS

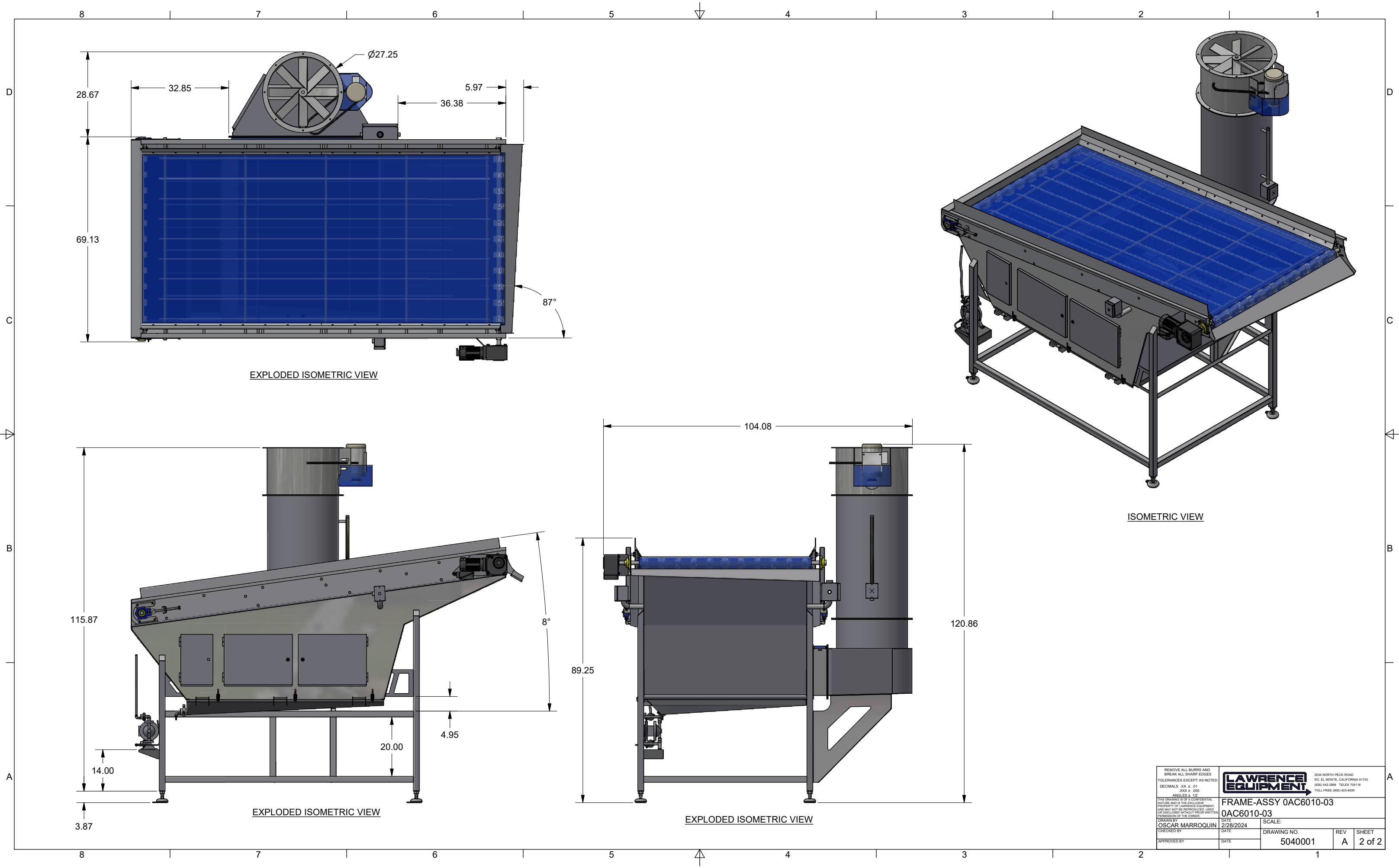
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MATERIALS LIST				
ITEM	FILE NAME	LE PART NUMBER	DESCRIPTION	QTY
1	270858-001.iam	270858-001	FRAME AMBIENT COOLER	1
2	BG013-07-P.ipt	BG013-07-P	SLIDE BRG 1.25 BORE SKWEZLOC	2
3	BG006-15-P.ipt	BG006-15	4 BLT FLNG BRG PLT 1.25 BORE	2
4	FA460025.ipt	FA460025	HEX NUT REG SS 1.00-8	4
5	MH020-12.ipt	MH020-12	VIB PAD MIG 1.0-8UNC X 4.0LG	4
6	360141-001.ipt	360141-001	CLEAT TRACK SPRKT 16TX1.50SQ	22
7	MH011-25.ipt	MH011-25	HEX LOCK TYPE LATCH	2
8	MH011-05.ipt	MH011-05	LATCH SS ACTION TYPE	3
9	301525-001.iam	301525-001	TAKE UP SCRW 11 CLNG CONV.	2
10	5040001-WS001-03-01.ipt	5040001-WS001-03-01	U CHANNEL HDP	84.447 ft
11	313392-003.ipt	313392-003	IDLE SHAFT/TAKEUP 1.25	1
12	5040001-MB258-01.ipt	5040001-MB258-01	BALANCE WEAVE 304 S/S 58 IN	24.583 ft
13	413377-001.ipt	413377-001	INFEED SLIDE/CHUTE 0AC3610	1
14	GM117-06.ipt	GM117-06	GEAR MOTOR .5HP 100-1 17RPM	1
15	270859-001.iam	270859-001	STACK FRAME SECTION - 0AC3610	1
16	270315-001.iam	270315-001	FILTER HOUSING-FRAME WELDMENT	1
17	412265-001.ipt	412265-001	FILTER CAP 0AC2410	1
18	313393-008.ipt	313393-008	DRIVE SHAFT 1.25 JOURNALS	1
19	412266-001.ipt	412266-001	TAKEUP BEARING BRACKET 0AC2410	4
20	FP004-02.ipt	FP004-02	GREASE FILTER BAFFLE SS TYPE-3	3
21	GT011-15.ipt	GT011-15	AXIAL FAN BELT DRIVE	1
22	412465-001.iam	412465-001	STACK EXTENSION 18IN STACK	1
23	301533.ipt	301533	LOCK COLLAR COOLING CONV.	4
24	MO102-52.ipt	MO102-52	MTR 2HP 1725RPM 145TC TEFC	1
25	422100-001.iam	422100-001	MOTOR GUARD	1
26	SA005-87.ipt	SA005-87	BUSHING H-TYPE 0.875 BORE	1
27	SA008-02.ipt	SA008-02	V BELT PULLEY SHEAVE	1
28	VB012-08.ipt	VB012-08	V-BELT SERIES 9003 B46	1
29	412267-002.ipt	412267-002	GEARBOX TORQUE ARM 1430-SUMI	1
30	306868-001.ipt	306868-001	SIDE SHIELD	4
31	SF010-24.ipt	SF010-24	GASKET VITON 2 INCH	1
32	414515-002.ipt	414515-002	INFEED CHUTE WEIGHT CONVEYOR	1
33	SF012-39.ipt	SF012-39	DUAL DIAPHRAM 0.5NPT PUMP	1
34	380440-001.iam	380440-001	ADAPT SF1.5 TO 0.25NPT	2
35	SF010-16.ipt	SF010-16	GASKET SILICONE 1.5	2
36	380442-001.iam	380442-001	ADAPT SF2.0 TO 0.25NPT	1
37	SF004-02.ipt	SF004-02	CLAMPS 2.0 OD/450 MAX PSI	1
38	PF40602-MF.ipt	PF40602-MF	ELBOW SS .25 90DEG MALE-FEMALE	3
39	HF004-33.ipt	HF004-33	BRS FITTING .37 HOSE .25 NPT	3
40	*Varies*	MH023-62	FOOD GRADE 0.375 X 0.50 HOSE	2
41	SF004-01.ipt	SF004-01	CLAMP 1.50 DIA 500 PSI MAX	2
42	407207-004.iam	407207-004	SCREEN FOR FRYER 34.1875X6.125	1
43	5040001-WS001-03-02.ipt	5040001-WS001-03-02	U CHANNEL HDP	65.685 ft
44	405817-001.iam	405817-001	SIDE EXTENTION ASPIRATION CONV	2

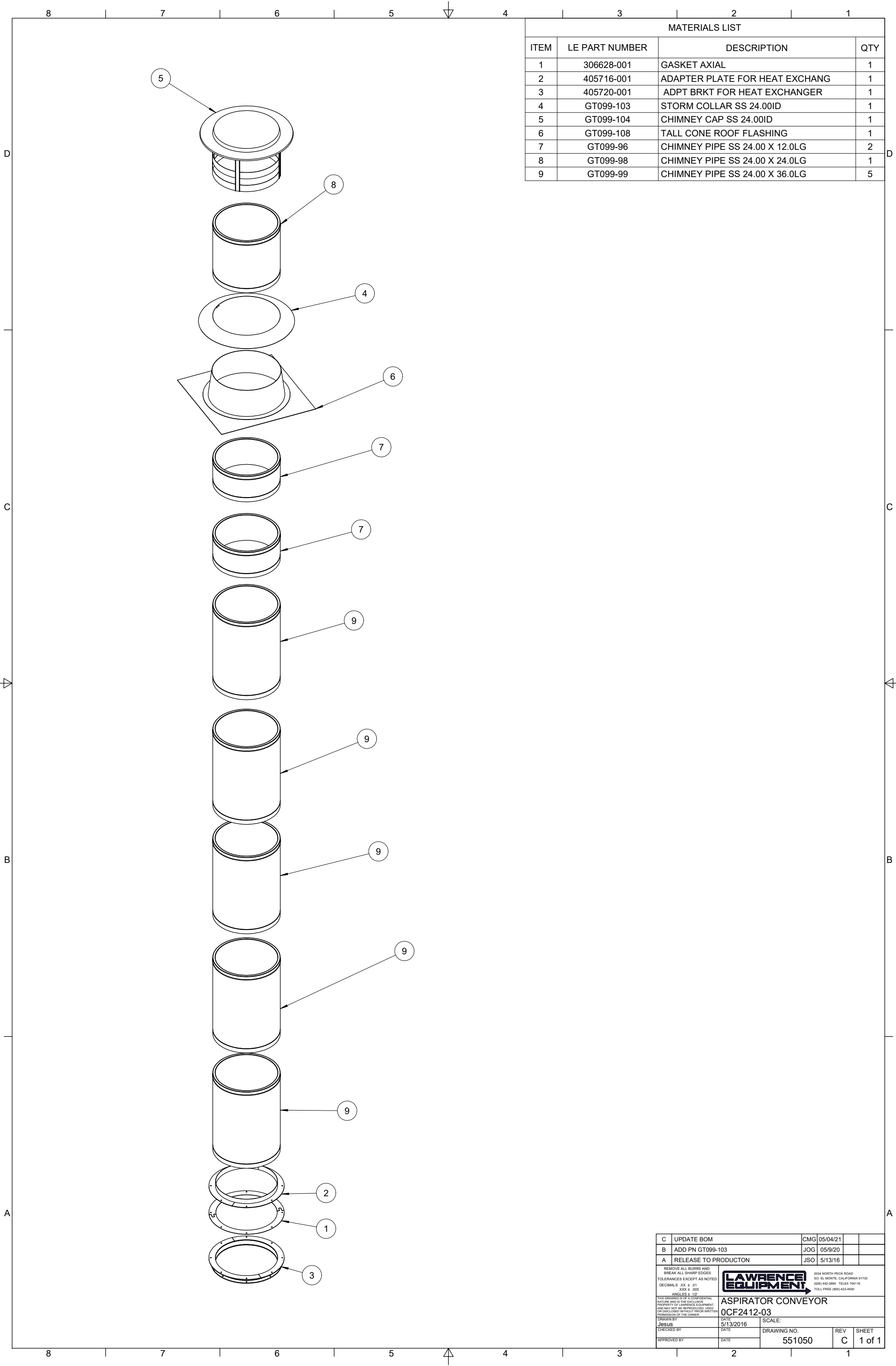


REMOVE ALL BURRS AND BREAK ALL SHARP EDGES. TOLERANCES EXCEPT AS NOTED DECIMALS .XX ± .01 XXX ± .005 ANGLES ± 1°		LAWRENCE EQUIPMENT		2034 NORTH PECK ROAD SO. EL MONTE, CALIFORNIA 91733 (626) 442-2884 TELEX 704116 TOLL FREE (800) 423-4000	
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DRAWN BY OSCAR MARROQUIN		DATE 2/28/2024		DRAWING NO. 5040001	
CHECKED BY		DATE		REV A	
APPROVED BY		DATE		SHEET 1 of 2	

A	RELEASE TO PRODUCTION	OKM	02/28/24		
NO	REVISION	BY	DATE	BY	DATE



REMOVE ALL BURRS AND BREAK ALL SHARP EDGES TOLERANCES EXCEPT AS NOTED DECIMALS .XX ± .01 XXX ± .005 ANGLES ± 1°		LAWRENCE EQUIPMENT		2034 NORTH PECK ROAD SO. EL MONTE, CALIFORNIA 91733 (626) 442-2894 TELEX 704116 TOLL FREE (800) 423-4000	
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DRAWN BY OSCAR MARROQUIN		DATE		DRAWING NO.	
CHECKED BY		DATE		REV	
APPROVED BY		DATE		SHEET	
		5040001		A 2 of 2	



MATERIALS LIST			
ITEM	LE PART NUMBER	DESCRIPTION	QTY
1	306628-001	GASKET AXIAL	1
2	405716-001	ADAPTER PLATE FOR HEAT EXCHANG	1
3	405720-001	ADPT BRKT FOR HEAT EXCHANGER	1
4	GT099-103	STORM COLLAR SS 24.00ID	1
5	GT099-104	CHIMNEY CAP SS 24.00ID	1
6	GT099-108	TALL CONE ROOF FLASHING	1
7	GT099-96	CHIMNEY PIPE SS 24.00 X 12.0LG	2
8	GT099-98	CHIMNEY PIPE SS 24.00 X 24.0LG	1
9	GT099-99	CHIMNEY PIPE SS 24.00 X 36.0LG	5

C	UPDATE BOM	CMG	05/04/21		
B	ADD PN GT099-103	JOG	05/9/20		
A	RELEASE TO PRODUCTION	JSO	5/13/16		
REMOVE ALL BURRS AND BREAK ALL SHARP EDGES TOLERANCES EXCEPT AS NOTED DECIMALS .XX ± .01 XXX ± .005 ANGLES ± 1/2° THIS DRAWING IS OF A CONFIDENTIAL NATURE AND IS THE EXCLUSIVE PROPERTY OF LAWRENCE EQUIPMENT, AND MAY NOT BE REPRODUCED, USED, OR DISCLOSED WITHOUT PRIOR WRITTEN PERMISSION OF THE OWNER.		LAWRENCE EQUIPMENT → 2034 NORTH PECK ROAD SO. EL MONTE, CALIFORNIA 91733 (626) 442-2894 TELEEX 704116 TOLL FREE (800) 423-4500			
		ASPIRATOR CONVEYOR OCF2412-03			
DRAWN BY Jesus	DATE 5/13/2016	SCALE:			
CHECKED BY	DATE	DRAWING NO. 551050	REV C	SHEET 1 of 1	
APPROVED BY	DATE				